



Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : V Semester-B. Tech Information Technology

Course Code & Title : IT3501 Full Stack Web Development

Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic** : Unit 3 / Connecting to MongoDB from Node.js
- **Course Outcome** : CO3
- **Topic Learning Outcome:** TLO14
- **Activity Chosen** : Reflection
- **Justification:**
 - Reflection is the method of assessing how well the students understand a particular concepts and identify their doubts about the particular topic.
 - connecting MongoDB with Node.js is a crucial technique for modern web application development. MongoDB is a NoSQL database, and its connection to Node.js allows developers to handle and manage data seamlessly.
 - This activity is chosen to make sure the students can understand how Node.js and MongoDB databases are connected.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - The activity was conducted at the end of the class after discussing about the steps involved in connecting MongoDB from Node.js with example program, for a total of 15 minutes as shown in Fig 1.
 - Node.js with MongoDB database connectivity coding questions were given for them to work out and discussed.
 - Students were asked to write their doubts or understanding level of the topic.
 - After writing explanations for the questions on the papers, the instructor distributed the paper to the students in the next class.
 - This activity helped the students to improve their understanding level.

• CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO5	PO10	PSO1
CO3	3	2	3	3	1	3

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO5	PO10	PSO1
Justification for correlation	Students will be able to understand the steps involved in connecting MongoDB with Node.js	Students will be able to analyse data storage and retrieval problems using MongoDB	Students will be able to develop modern web application to solve a real-world problem	Students will be able to utilize MongoDB for modern database management	Students will be able to communicate effectively the MongoDB connectivity to team members	Students will be able to use MongoDB capabilities in handling large datasets.

• Images / Screenshot of the practice:



Fig 1: Reflection Activity

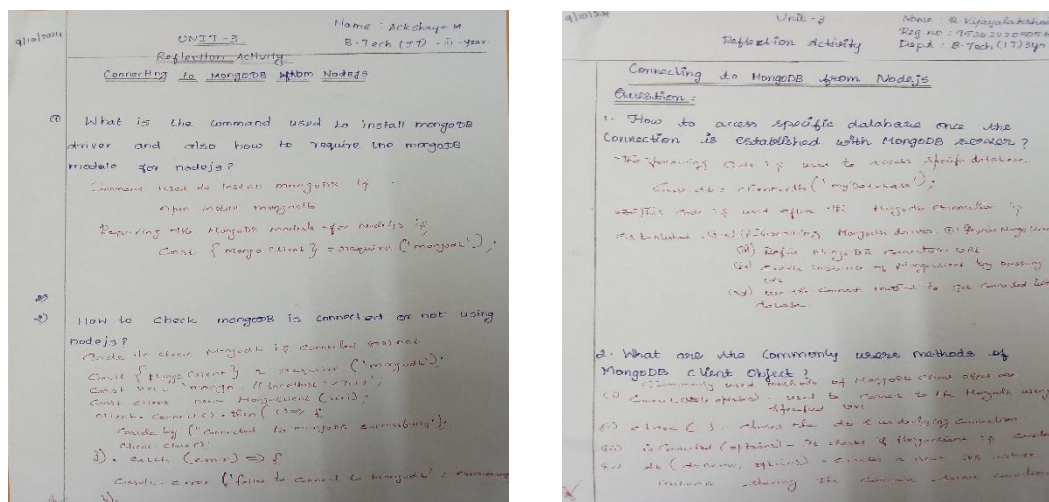


Fig 2: Sample Sheet of Reflection Activity

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The Students were allowed to evaluate themselves and identify the steps in connecting MongoDB with node.js where they needed further clarification.
- Students thought that the activity allowed them to ask their doubts and get clarified from instructor.

- ❖ ***Benefit of the practice:***

- This activity makes it simple to identify students who didn't understand the MongoDB database connectivity with Node.js and to provide them with further explanations with example program, so raising their understanding level.

- ❖ ***Challenges faced in implementation:***

- Some students found error handling techniques in MongoDB connection slightly challenging.
- Few students opted not to ask any questions, thus I was unable to really get an understanding of all the students' perspectives.

In the next class, I answered all of the students' questions and all doubts were clarified.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/RE_C_BV_22-23_U%20III.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1.KVS_Reflection_FDS.pdf

Signature of Faculty Member

HOD